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1. Current Status of URDF Components in Hydro:

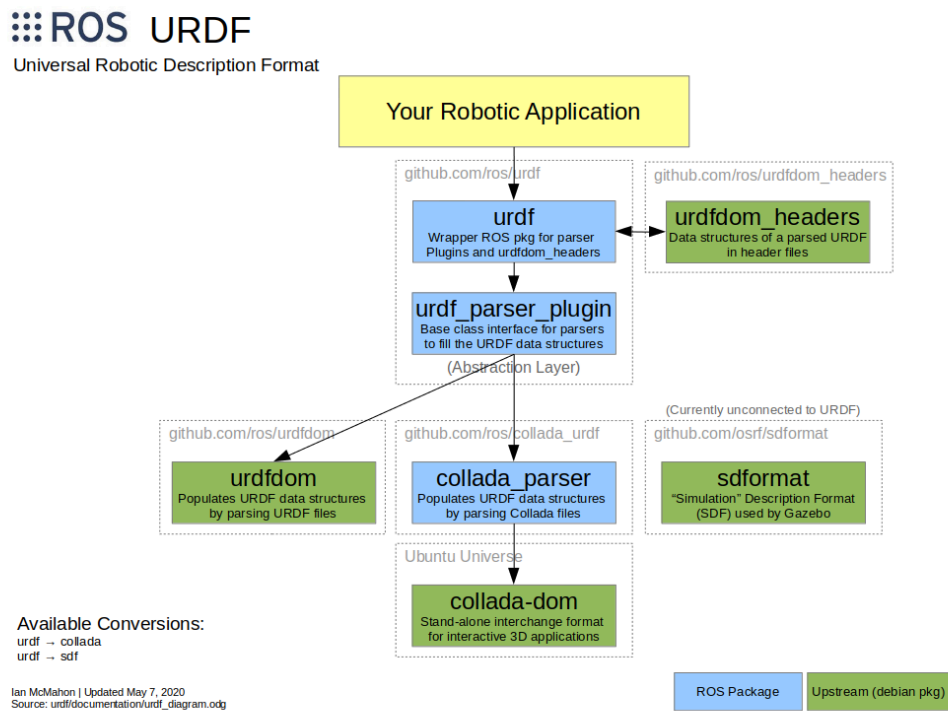
1. Overview

This package contains a number of XML specifications (/urdf/XML) for robot models, sensors, scenes, etc. Each XML specification has a corresponding parser in one or more languages.

	XML Description	C++	Python	XML Schema
Model	model (/urdf/XML/model)	Parser API http://www.ros.org/doc/api/urdf/html/ Model API https://github.com/ros/urdfdom_headers/tree/master/include/urdf_model	Model API http://wiki.ros.org/urdfdom_py	urdf.xsd https://raw.githubusercontent.com
Sensor	sensor (/urdf/XML/sensor)			

2. Components

New in Hydro A number of different packages and components make up urdf. The following diagram attempts to explain the relationship between these components:



2.1 Current Status of URDF Components in Hydro:

- The core URDF parser and data structures (urdfdom, urdfdom_headers) have been pushed upstream into stand alone (no ROS-dependencies) software packages that will in the future be released into Ubuntu completely separate from ROS.
- There is now a distinction between a URDF file and a URDF data structure. A URDF file follows the XML format as described on the ros.org wiki. A URDF data structure is a set of generic classes that various formats (currently URDF and Collada) can be parsed into.
- A new plugin abstraction layer (urdf_parser_plugin) allows the URDF data structures to be populated with various file formats (currently URDF and Collada)
- The ROS packages urdf_parser and urdf_interface have been deprecated in Groovy and removed in Hydro.

3. Getting Started

There is large set of **tutorials** on how to build up your own robot models using the URDF specification. Check out the [urdf/Tutorials \(/urdf/Tutorials\)](/urdf/Tutorials) page.

We also developed a macro language called xacro ([/xacro](#)) to make it easier to maintain the robot description files, increase their readability, and to avoid duplication in the robot description files.

4. Examples

See this page (</urdf/Examples>) for a list of robots described by a URDF model.

5. Tools

5.1 Verification

A command line tool `check_urdf` attempts to parse a file as a URDF description, and either prints a description of the resulting kinematic chain, or an error message.

For example, to run this tool on the `pr2` urdf, first create the urdf file by running:

```
roslaunch xacro xacro.py `rospack find pr2_description`/robots/pr2.urdf.xacro -o /tmp/pr2.urdf
```

New in Indigo

Then run the check by running:

Note: You may need to run `sudo apt-get install liburdfdom-tools`.

```
check_urdf pr2.urdf
```

New in Hydro

Then run the check by running:

```
roslaunch urdfdom check_urdf /tmp/pr2.urdf
```

and you should see something resembling:

```
robot name is: pr2
----- Successfully Parsed XML -----
root Link: base_footprint has 1 child(ren)
  child(1): base_link
    child(1): base_laser_link
    child(2): bl_caster_rotation_link
      child(1): bl_caster_l_wheel_link
      child(2): bl_caster_r_wheel_link
    child(3): br_caster_rotation_link
      child(1): br_caster_l_wheel_link
      child(2): br_caster_r_wheel_link
    child(4): fl_caster_rotation_link
      child(1): fl_caster_l_wheel_link
      child(2): fl_caster_r_wheel_link
    child(5): fr_caster_rotation_link
      child(1): fr_caster_l_wheel_link
      child(2): fr_caster_r_wheel_link
    child(6): torso_lift_link
      child(1): head_pan_link
        child(1): head_tilt_link
          child(1): head_plate_frame
            child(1): sensor_mount_link
              child(1): double_stereo_link
                child(1): narrow_stereo_link
      ...
```

5.2 Visualization

Visual Studio Code has a [ROS extension \(https://hiro-group.ronc.one/vscode_urdf_previewer.html\)](https://hiro-group.ronc.one/vscode_urdf_previewer.html) that supports URDF visualization.

To get a graphviz diagram of your urdf file, do the following:

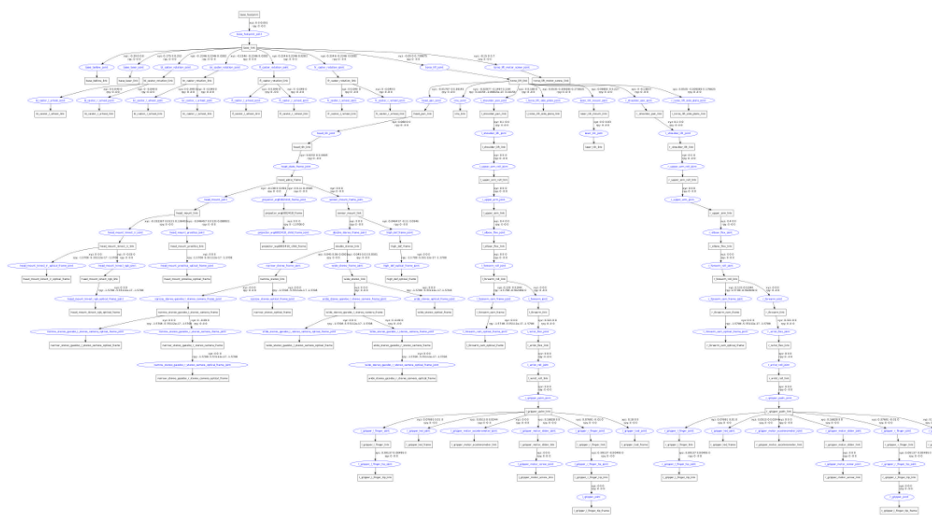
New in Indigo In Indigo, the `urdf_to_graphviz` tool has moved to the `liburdfdom-tools` package. You may need to run `sudo apt-get install liburdfdom-tools` if you can't use `urdf_to_graphviz`

```
urdf_to_graphviz pr2.urdf
```

New in Hydro

```
urdf_to_graphviz pr2.urdf
```

The result is a file called pr2.pdf that looks something like this:



5.3 Creating URDF

URDF is XML, which is a text format. So you can create the files in any text editor. An IDE with good XML support is recommended.

If there already exists a CAD model of the robot, there are some tools that can be used to convert the CAD model to URDF. Here is a non-exhaustive list:

- PTC Creo to URDF (iCub) (<https://github.com/icub-tech-iiit/cad-libraries/wiki/Prepare-PTC-Creo-Mechanism-for-URDF>)
- SolidWorks URDF exporter (http://wiki.ros.org/sw_urdf_exporter)
- fusion2urdf (<https://github.com/syuntoku14/fusion2urdf>)
- OnShape to Robot (<https://onshape-to-robot.readthedocs.io/en/latest/>)
- Phobos Blender plugin (<https://github.com/dfki-ric/phobos>)

6. Known Issues

- Sensor model information is not included (except in gazebo extensions for simulated sensors).

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